

American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

PROJECT TITLE : CareLink - Online Emergency Ambulance and Hospital Booking System

Semester: Spring 25-26

Group:		***Section:***
SL	**Student Name**	**Student ID**
1	MD.Rafayet Hossain Zanith	23-52178-2
2	Samiya Siddiqua Sympa	23-52240-2
3	Aminul Islam Raju	23-50969-1
4	Ishrak Jahan Mim	23-53017-3
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1. PROJECT PROPOSAL

1.1 Background to the Problem

Healthcare emergencies require immediate medical attention and quick access to hospital facilities. In many developing countries, including Bangladesh, patients and their families often face serious difficulties when trying to access hospital services during emergency situations. One of the major problems is the lack of real-time information regarding hospital resources such as available beds, ICU beds, wards, and private rooms. As a result, patients frequently travel from one hospital to another searching for available treatment facilities. This delay can critically affect the patient's health condition and in some cases may even lead to death before proper medical care is received.

Another significant challenge is the availability of emergency transportation. During emergencies, patients and their families often struggle to find an ambulance quickly. Traditional methods such as calling multiple ambulance services or physically visiting hospitals waste valuable time. Many ambulance services are not centrally managed, and there is no integrated system that allows patients to quickly request an ambulance and track its availability.

Furthermore, hospitals currently manage bed availability and patient records through manual or isolated systems. Because of this, there is no centralized platform that allows patients to check hospital capacity before arriving. This lack of transparency creates confusion, overcrowding, and inefficient utilization of healthcare resources.

To address these issues, this project proposes the development of an CareLink - Online Emergency Ambulance and Hospital Booking System. The system will provide a digital platform where patients can quickly find nearby hospitals, check real-time availability of hospital resources such as beds, ICU beds, wards, and rooms, and book them in advance. In addition, the system will allow users to request an ambulance directly from their location to transport them to the selected hospital.

The proposed system will integrate three major stakeholders: patients, ambulance drivers, and hospital management. Patients will be able to search for hospitals based on location and available facilities, book hospital beds, request ambulance services, and manage their medical bookings through the platform. Ambulance drivers will receive emergency requests from patients and will have the option to accept or reject these requests depending on their availability. Hospital management will have administrative control over the system to update hospital resource availability, manage patient bookings, generate bills, and coordinate with ambulance drivers.

Overall, this system seeks to improve emergency healthcare accessibility, reduce patient suffering during critical situations, and enhance the efficiency of hospital resource management. The project aims to create a practical technological solution that addresses real-life healthcare challenges in Bangladesh.

1.2 Selection of Process Model

For the development of CareLink, the team has selected the **Agile Software Development Model**, specifically implementing the **Scrum framework**. Agile methods are considered lightweight, people-based approaches where the entire software is not developed at once. Instead, development is broken into short iterative cycles, each spanning requirements analysis, design, implementation, and testing. A key characteristic of Agile is that iteration lengths are kept between one to four weeks and must not exceed thirty days. Research has shown that shorter iterations lead to lower complexity and risk, better feedback, and higher productivity compared to traditional long-cycle models.

Agile was chosen for CareLink for several reasons. The system consists of multiple independent modules including the Patient Panel, Ambulance Driver Panel, and Hospital Admin Panel, each of which can be developed and tested in separate sprints without waiting for the entire system to be complete. Additionally, since requirements may be refined throughout the semester based on faculty feedback. Agile provides the flexibility to accommodate changes without disrupting the overall project plan which a rigid plan-driven model like Waterfall could not offer.

Within Agile, the team specifically adopted the **Scrum framework**. The Scrum process consists of three phases. The **Pre-game phase** involves creating a Product Backlog of all known requirements, prioritizing them, estimating effort, and planning the high-level system architecture. The **Development phase** consists of time-boxed Sprints where the system is built incrementally. Each Sprint for CareLink is two weeks long and delivers a working portion of the system. The **Post-game phase** covers final integration, system testing, and documentation preparation for submission.

The Scrum framework also defines clear roles. The **Scrum Master** ensures the project follows Scrum rules and progresses as planned. The **Product Owner** manages the Product Backlog and makes decisions on task priorities. The **Scrum Team** organizes itself to achieve sprint goals, create the Sprint Backlog, and estimate effort. These roles were distributed among the four CareLink team members to ensure shared responsibility. Key Scrum practices followed included Sprint Planning Meetings at the start of each sprint. Daily Scrum meetings to track progress and resolve blockers, and Sprint Review Meetings at the end of each sprint to present completed features and collect feedback.

Overall the Agile Scrum model was the most suitable choice for CareLink because it supports iterative delivery, encourages team collaboration, accommodates changing requirements, and allows continuous testing and improvement throughout the development lifecycle.

2. Timeline Chart / Gantt Chart

CareLink – Online Emergency Ambulance & Hospital Booking System												
Phase	**Task**	**Week 1**	**Week 2**	**Week 3**	**Week 4**	**Week 5**	**Week 6**	**Week 7**	**Week 8**	**Week 9**	**Week 10**	**Week 11**	**Week 12**
Phase 1: Documentation	Write Project Proposal	✓											
	Draw Use Case Diagram	✓											
	Write SRS / Requirements		✓										
	Draw Class & ER Diagram		✓										
Phase 2: UI/UX Design	Design Auth Screens (Figma)			✓									
	Design Patient Screens (Figma)			✓									
	Design Driver & Admin Screens				✓								
Phase 3: Backend Setup	Set Up Server & Database				✓								
	Implement JWT Auth & RBAC					✓							
	Set Up API Route Structure					✓							
Phase 4: Patient Module	Registration & Login API						✓						
	Hospital Search & Booking API						✓						
	Ambulance Request & Tracking							✓					
	Payment & Notification Module							✓					
Phase 5: Driver Module	Driver Dashboard & Requests							✓					
	Live Location & Trip History								✓				
Phase 6: Admin Module	Admin Dashboard & Bed Mgmt								✓				
	Booking & Billing Management									✓			
	Notifications & Reports									✓			
Phase 7: Integration & Test	Module Integration										✓		
	Write & Execute Test Cases										✓		
	Bug Fixing & Re-Testing											✓	
Phase 8: Final Submission	Complete Project Report											✓	
	Prepare Project Demo												✓
	Final Submission												✓
Legend:		✓ Completed			● Upcoming / In Progress								

3. SOFTWARE REQUIREMENTS SPECIFICATIONS (SRS) / PRODUCT REQUIREMENTS DOCUMENT (PRD)

3.1 Scopes and Features

Scope 1 : Patient-Facing Core System

The Patient-Facing Core System is a web-based application developed to automate and manage the hospital bed and resource booking process for patients. The system is designed to support secure user authentication with role-based access for patients, ensuring that each patient can access only their own booking data and functionalities. It enables patients to search for nearby hospitals by location, view detailed hospital profiles, check real-time availability of beds, ICU units, cabins, and wards, book desired hospital resources, cancel existing bookings, view complete booking history, and receive notifications regarding booking confirmations, cancellations, and reminders. The system includes validation mechanisms to prevent duplicate or invalid bookings, error handling for failed operations, and security controls to protect patient data. The scope of the system focuses on reducing the time patients spend searching for hospital resources, eliminating manual hospital visits, and providing a smooth and secure booking experience, while excluding ambulance tracking, online payments, bill generation, driver functions, and administrative backend features.

Scope 2 : Emergency Ambulance & Live Tracking Module

The Emergency Ambulance & Live Tracking Module is a web-based application developed to automate and manage real-time ambulance request, dispatch, and location tracking between patients and ambulance drivers. The system is designed to support secure user authentication with role-based access for patients and ambulance drivers, ensuring that each user can access only the functionalities relevant to their role. It enables patients to request an ambulance with pickup location details and track the ambulance live on a map, while allowing drivers to register, log in, view incoming requests, accept or reject trips based on availability, share live location during active trips, update their availability status, and view trip history. The system includes validation mechanisms to prevent multiple drivers from accepting the same request, error handling for cases where no driver accepts a request, and security controls to ensure only authorized drivers receive requests. The scope of the system focuses on improving emergency response times, reducing patient transportation delays, and providing reliable ambulance coordination, while excluding hospital bed booking, online payments, bill generation, hospital admin functions, and patient booking history.

Scope 3 : Admin & Billing Management System

The Admin & Billing Management System is a web-based application developed to automate and manage hospital administration, patient booking oversight, and financial transactions. The system is designed to support secure user authentication with role-based access for hospital administrators, ensuring that each admin can access only the functionalities relevant to their assigned hospital. It enables administrators to update real-time availability of beds, ICU units, cabins, and wards, view and manage all incoming patient bookings, confirm or cancel booking requests, generate bills for patients after discharge or service completion, update billing information when corrections are needed, view ambulance request logs, manage hospital profile details, monitor current bed occupancy, and send notifications to patients regarding admission, discharge, payment, or booking status. The system includes payment handling for online transactions, invoice generation for patient records, validation mechanisms to ensure billing accuracy, error handling for payment failures, and security controls to prevent unauthorized administrative actions. The scope of the system focuses on improving operational efficiency, reducing manual errors in billing and resource management, and providing a smooth and secure administrative experience, while excluding ambulance driver functions, live location tracking, patient-side search features, third-party payment gateway integrations, mobile applications, and offline transaction processing.

3.2 User Story Table

User Story Table

CareLink-Online Emergency Ambulance & Hospital Booking System

1. Patient

As A/An	I want to	So that	Acceptance Criteria
Patient	Register / Login	Access system services	Valid credentials required; redirect to dashboard
Patient	Search nearby hospitals	Find closest hospital	System displays hospitals based on location
Patient	View hospital details	Compare services	Details must load correctly

Patient	Check bed / ICU availability	Choose available hospital	Real-time availability displayed
Patient	Book bed / ICU / cabin / ward	Ensure treatment	Booking saved and confirmation shown
Patient	Cancel booking	Modify plans	Booking removed and updated in system
Patient	View booking history	Track records	All previousbookings displayed
Patient	Request ambulance	Reach hospital quickly	Request sent to available drivers
Patient	Track ambulance live location	Know arrival time	Live location updates in real-time
Patient	Make online payment	Complete booking	Payment validated and confirmed
Patient	View bill	Verify charges	Bill details displayed correctly
Patient	Receive notifications	Stay updated	Notifications delivered instantly

2. Ambulance Driver

As A/An	I want to	So that	Acceptance Criteria
Ambulance Driver	Register / Login	Access system	Valid login redirects to dashboard
Ambulance Driver	View incoming requests	See patient requests	Requests appear instantly
Ambulance Driver	Accept / Reject request	Manage workload	Status updates immediately
Ambulance Driver	Share live location	Patient can track	Location updates in real-time

Ambulance Driver	Update availability status	Show availability	Status reflected in system
Ambulance Driver	View trip history	Track trips	Trip records stored and displayed

3. Hospital Admin

As A/An	I want to	So that	Acceptance Criteria
Hospital Admin	Login	Manage system	Valid credentials required
Hospital Admin	Update bed / ICU availability	Provide real-time data	Updates reflected instantly
Hospital Admin	View & manage patient bookings	Organize resources	Booking list displayed correctly
Hospital Admin	Confirm / cancel bookings	Control capacity	Status updated and patient notified
Hospital Admin	Generate bill for patient	Handle payments	Bill generated and stored
Hospital Admin	Update billing information	Maintain accuracy	Changes saved immediately
Hospital Admin	View ambulance request logs	Monitor services	Logs displayed correctly
Hospital Admin	Manage hospital profile	Keep info updated	Changes saved in system
Hospital Admin	Monitor bed occupancy	Track usage	Dashboard shows accurate data
Hospital Admin	Send notifications to patients	Inform patients	Notifications delivered successfully

3.3 Requirements Traceability Matrix

3.3.1 Functional Requirements

FR-1: User Registration The system shall allow patients to create an account by providing required information such as name, phone number, address, and password.

FR-2: User Login The system shall allow registered users (Patient, Driver, Admin) to log in using a valid username/phone and password based on their assigned role.

FR-3: Role-Based Access Control The system shall restrict system features and dashboards according to the authenticated user's role (Patient, Driver, or Admin).

FR-4: Password Security The system shall mask passwords during input and enforce password complexity rules during registration.

FR-5: Search Nearby Hospitals The system shall allow patients to search for nearby hospitals based on their current GPS location or manual address input.

FR-6: Real-Time Bed & ICU Availability The system shall display real-time bed and ICU availability for each hospital, updated instantly when bookings or admin updates occur.

FR-7: Hospital Bed Booking The system shall allow patients to book an available hospital bed, storing the booking information in the database with a unique booking ID.

FR-8: Ambulance Request The system shall allow patients to request an ambulance, sending the request instantly to available drivers.

FR-9: Real-Time Ambulance Tracking The system shall allow patients to track the assigned ambulance in real-time, showing location, ETA, and driver status.

FR-10: View Booking History The system shall allow patients to view their complete booking history, including past and active bookings.

FR-11: Cancel Booking The system shall allow patients to cancel an existing booking with a confirmation prompt before final cancellation.

FR-12: Online Payment The system shall allow patients to make online payments for hospital bookings and ambulance services using available payment methods such as card or mobile banking. A payment confirmation and receipt shall be generated after each successful transaction.

FR-13: View Bill The system shall allow patients to view their generated bills including all service charges, ambulance charges, and total amount due.

FR-14: Driver Request Notification The system shall display incoming ambulance requests instantly to all available drivers.

FR-15: Accept or Reject Requests The system shall allow drivers to accept or reject ambulance requests, updating the request status immediately in the system.

FR-16: View Patient Location The system shall show the patient's pickup location using accurate GPS data to the assigned driver and allow the driver to navigate to the patient using the provided location.

FR-17: Driver Availability Toggle The system shall allow drivers to update their availability status (Online/Offline/Busy), which affects which requests they receive.

FR-18: Complete Trip The system shall allow drivers to mark a trip as completed, saving all trip details such as distance, time, and route to the database.

FR-19: Admin Secure Login The system shall allow hospital admins to log in securely with valid credentials to access the admin dashboard.

FR-20: Update Bed & ICU Availability The system shall allow admins to update bed and ICU availability numbers in real time, instantly reflecting changes to patients.

FR-21: Manage Bookings The system shall allow admins to view, filter, and manage all patient bookings from a centralized list.

FR-22: Approve or Reject Bookings The system shall allow admins to approve or reject patient booking requests, automatically notifying the patient of the decision.

FR-23: Generate Bill The system shall allow admins to generate bills for patient treatment, including all services and charges.

FR-24: Occupancy Dashboard The system shall provide admins with a dashboard showing current hospital occupancy, bed usage, and ambulance status.

FR-25: Logout The system shall allow all users to log out successfully, clearing their session and preventing unauthorized access.

FR-26: Input Validation The system shall display clear validation and error messages for invalid inputs without refreshing the page

3.3.2 Non-Functional Requirements

NFR-1: Performance The system shall respond to user actions such as hospital search, bed booking, and ambulance tracking within 3 seconds under normal operating conditions.

NFR-2: Request Speed The system shall display ambulance requests to available drivers within 2 seconds of patient submission.

NFR-3: Availability Update Speed The system shall update bed and ICU availability within 1 second after an admin change or booking.

NFR-4: Concurrent Users The system shall support at least 100 concurrent users including patients, drivers, and admins without performance degradation.

NFR-5: Reliability & Availability The system shall be available 99% of the time, excluding scheduled maintenance windows.

NFR-6: Fault Tolerance In the event of a minor system failure, the system shall recover automatically without any data loss.

NFR-7: Data Backup All booking, patient, trip, and billing data shall be backed up daily to prevent data loss.

NFR-8: Security — Encryption All sensitive data including passwords and patient information shall be encrypted during storage and transmission.

NFR-9: Session Management User sessions shall expire after 30 minutes of inactivity to ensure security while allowing sufficient time for users during emergency situations.

NFR-10: Access Restriction The system shall prevent unauthorized users from accessing protected features and pages using role-based access control.

NFR-11: Input Validation Security All inputs shall be validated to prevent SQL injection, XSS attacks, and other malicious entries.

NFR-12: Usability The system shall be easy to use, allowing users with basic smartphone or computer skills to navigate and complete tasks without prior training.

NFR-13: Interface Consistency All pages, forms, and messages shall maintain a uniform and intuitive design across patient, driver, and admin panels.

NFR-14: Mobile Responsiveness The system shall be fully responsive and compatible with mobile devices, as the majority of users in Bangladesh are expected to access the system through smartphones.

NFR-15: GPS Accuracy The system shall provide GPS location accuracy within 50 meters for ambulance tracking and hospital search.

NFR-16: Real-Time Refresh Rate Real-time ambulance location updates shall refresh every 5 seconds or less.

NFR-17: Scalability The system shall be designed to handle future growth including more hospitals, ambulances, users, and cities.

NFR-18: Maintainability The system shall be developed using modular and reusable code to allow easy updates and bug fixes.

NFR-19: Documentation Technical documentation shall be provided for all modules including database schema, APIs, and system workflows.

4. SOFTWARE DESIGN

4.1 System Design:

Use Case Diagram

The Use Case Diagram for this system includes three actors: Patient, Ambulance Driver, and Hospital Admin. All 28 use cases are enclosed within the system boundary. The diagram includes include and extend relationships between use cases to show dependencies and optional interactions.

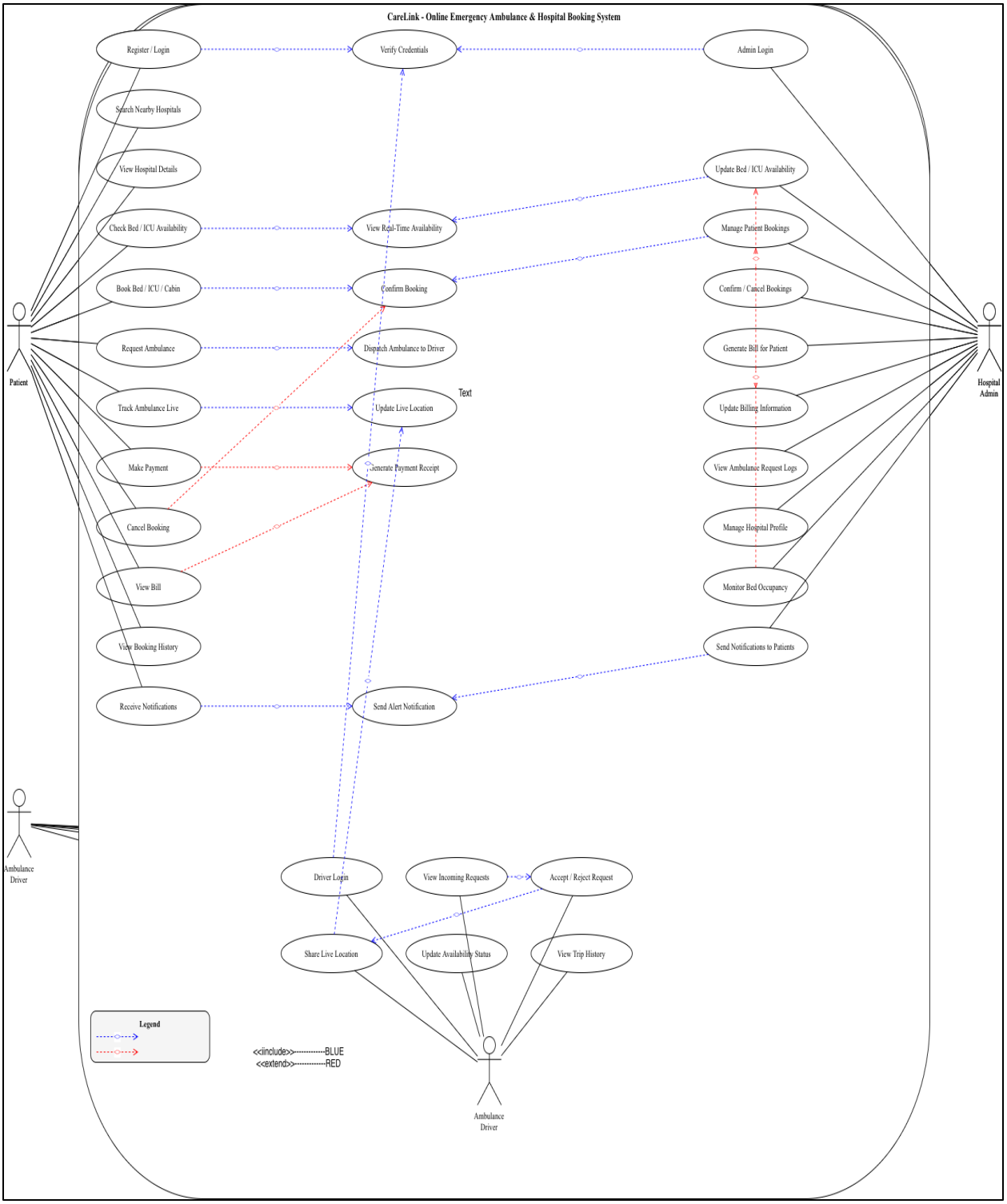
The Patient actor interacts with 12 use cases including Login/Register, Search Nearby Hospitals, View Hospital Details, Check Bed/ICU Availability, Book Bed/ICU/Cabin, Request Ambulance, Track Ambulance Live, Make Payment, Cancel Booking, View Bill, View Booking History, and Receive Notifications.

The Ambulance Driver actor interacts with 6 use cases including Driver Login, View Incoming Requests, Accept/Reject Request, Share Live Location, Update Availability Status, and View Trip History.

The Hospital Admin actor interacts with 10 use cases including Admin Login, Update Bed/ICU Availability, Manage Patient Bookings, Confirm/Cancel Bookings, Generate Bill for Patient, Update Billing Information, View Ambulance Request Logs, Manage Hospital Profile, Monitor Bed Occupancy, and Send Notifications to Patients.

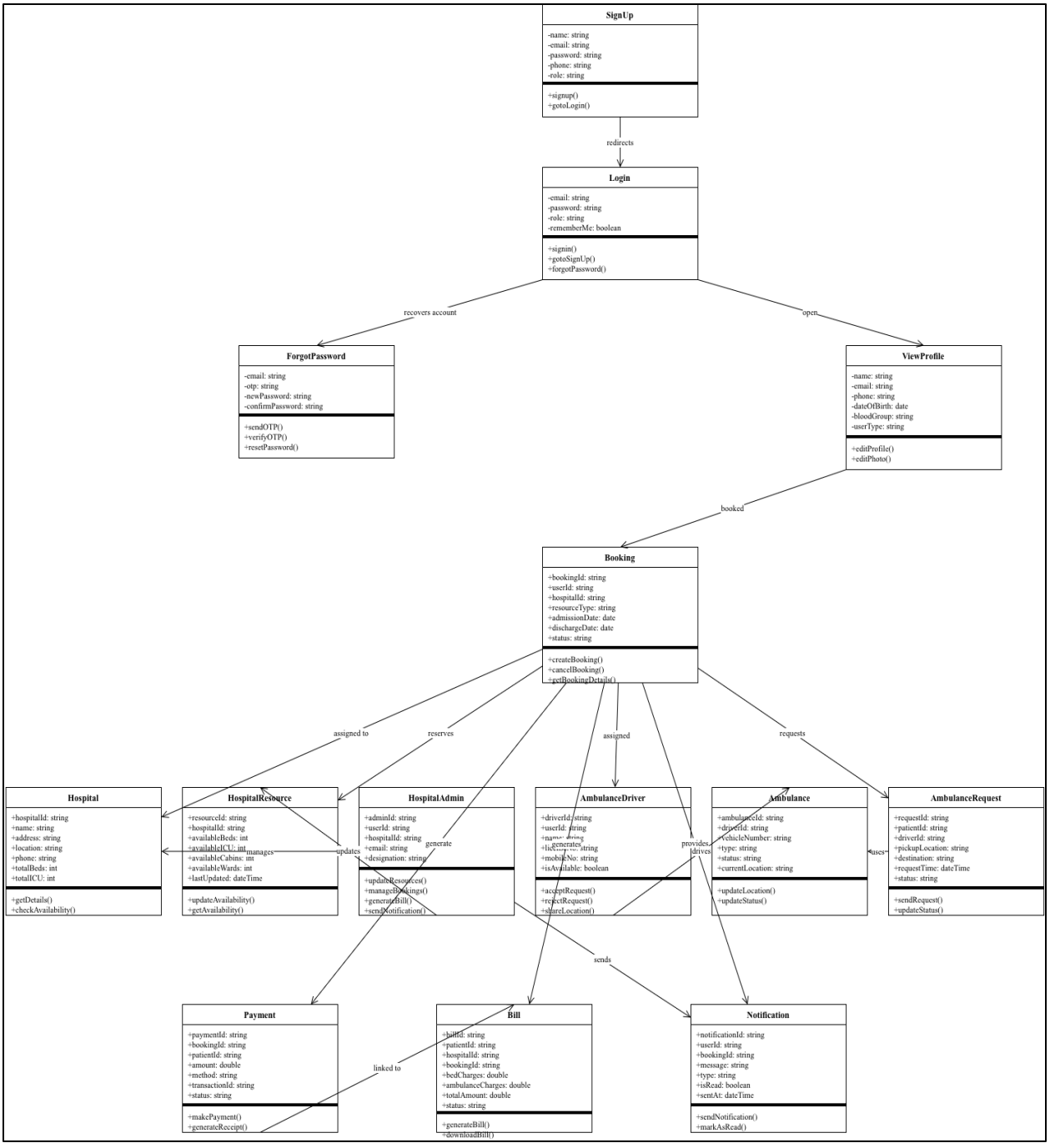
Include relationships exist between Login and Verify Credentials, Book Bed and Confirm Booking, Request Ambulance and Dispatch Ambulance to Driver, and Track Ambulance and Update Live

Location. Extend relationships exist between Make Payment and Generate Payment Receipt, View Bill and Generate Payment Receipt, and Send Notifications and Receive Notifications.

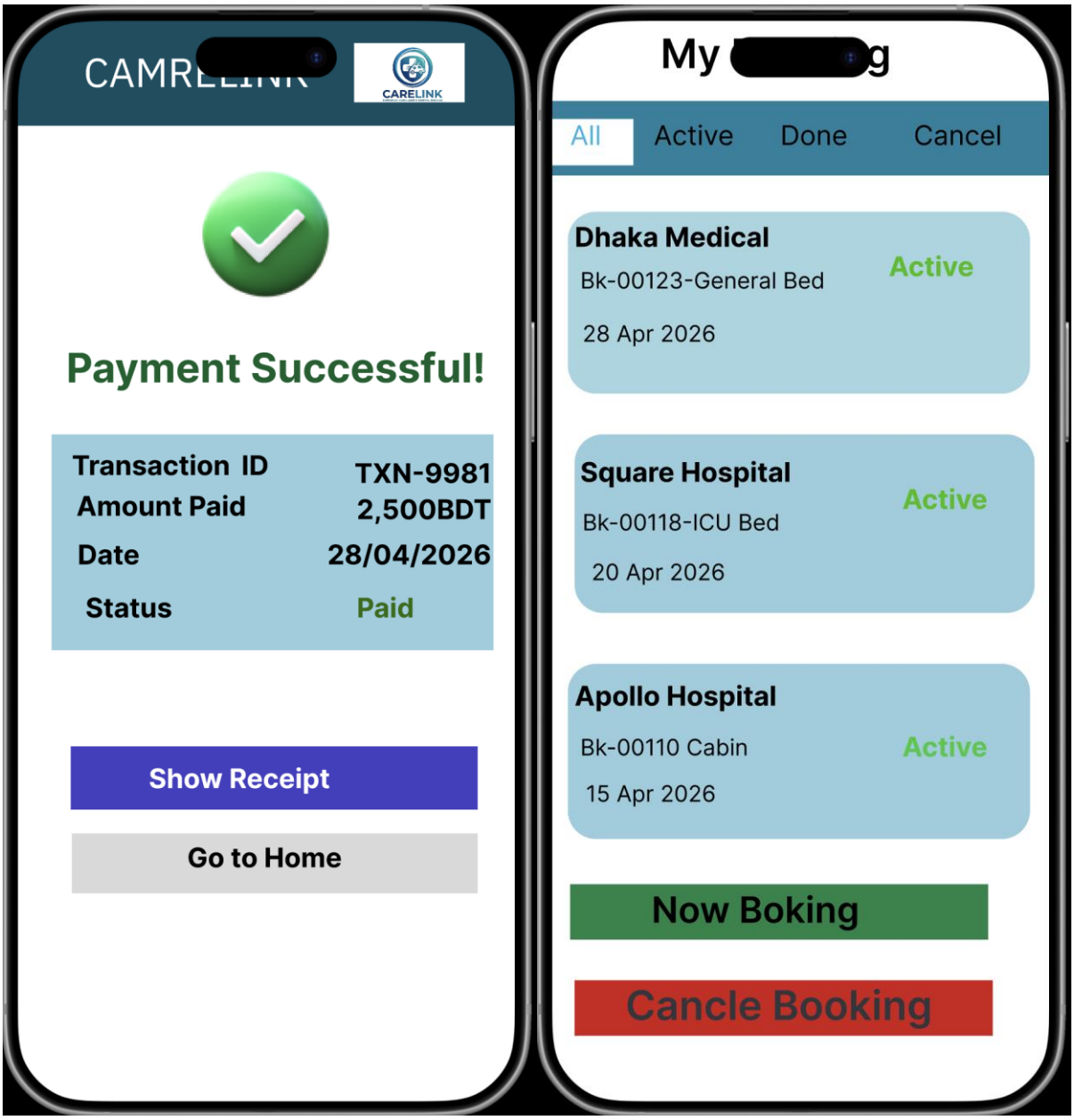


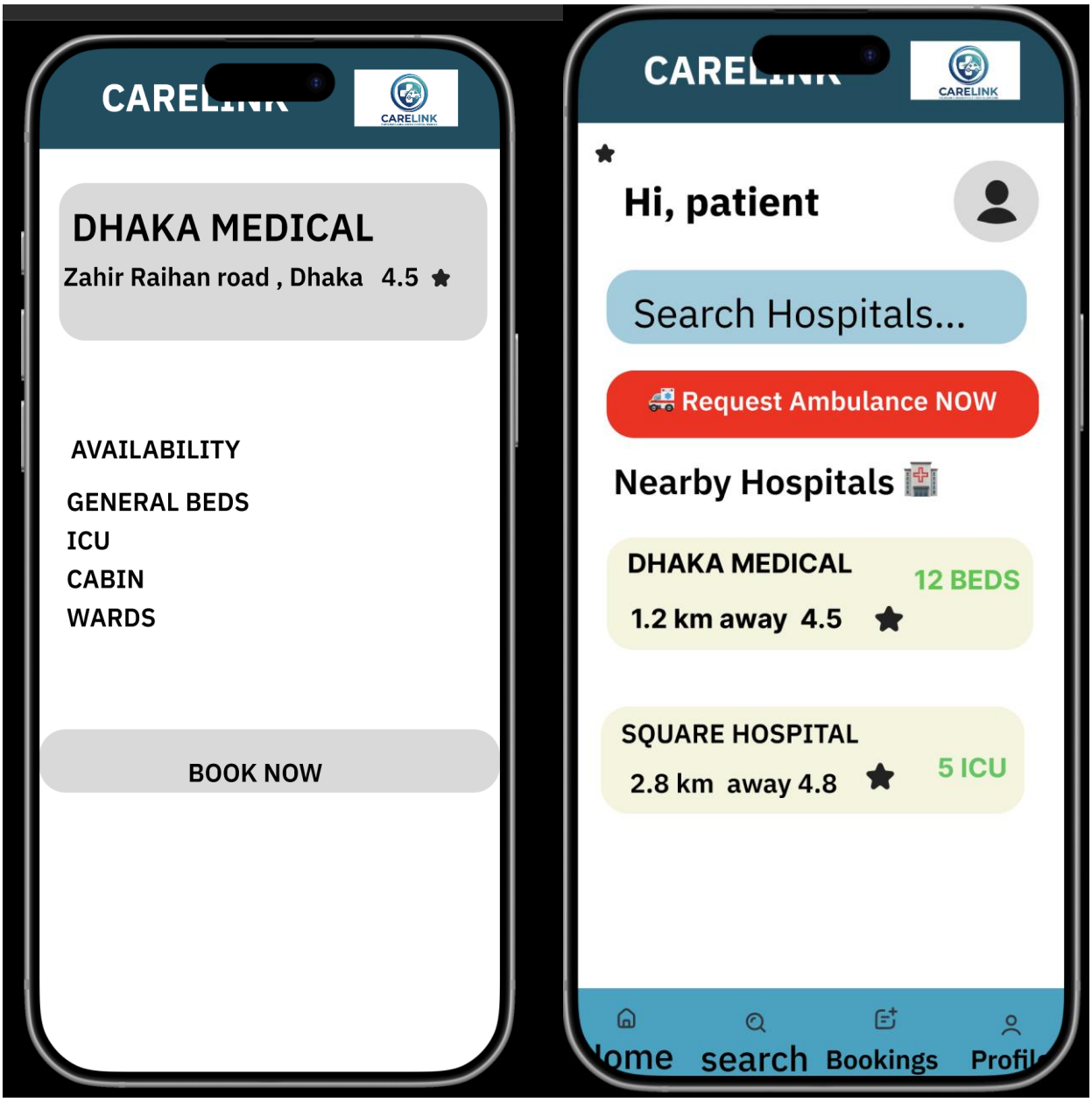
Class Diagram

The Class Diagram defines the main entities (classes) of the system, their attributes, methods, and relationships. The system consists of 14 main classes:

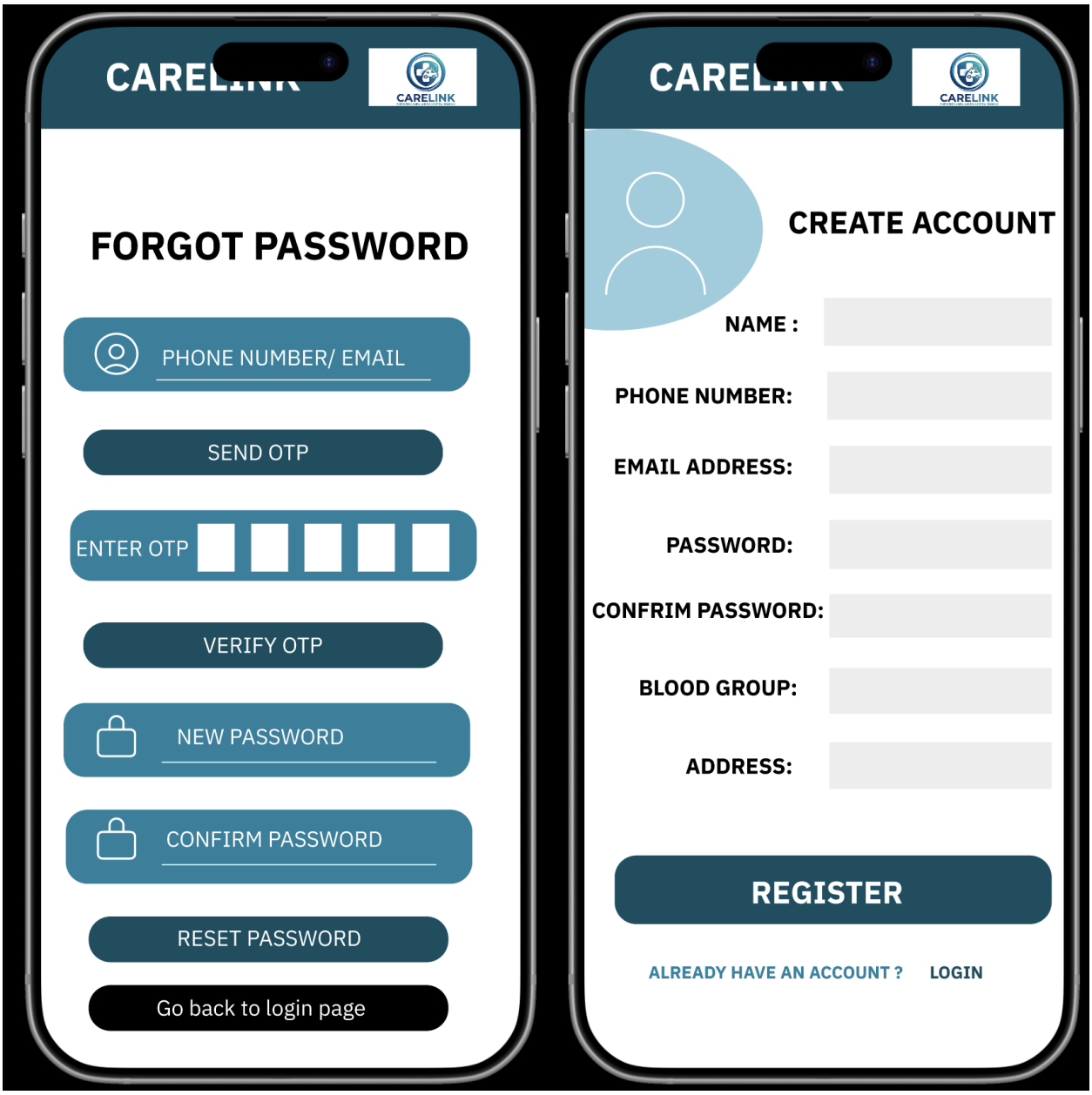


3.1 UI / Wireframe Design using Figma


CAMRELINK
CARELINK
Payment Successful!
Transaction ID
TXN-9981
Amount Paid
2,500BDT
Date
28/04/2026
Status
Paid
Show Receipt
Go to Home
My g
All
Active
Done
Cancel
Dhaka Medical
Bk-00123-General Bed
Active
28 Apr 2026
Square Hospital
Bk-00118-ICU Bed
Active
20 Apr 2026
Apollo Hospital
Bk-00110 Cabin
Active
15 Apr 2026
Now Boking
Cancle Booking


CARELINK
DHAKA MEDICAL
Zahir Raihan road , Dhaka 4.5
AVAILABILITY
GENERAL BEDS
ICU
CABIN
WARDS
BOOK NOW
CARELINK
Hi, patient
Search Hospitals...
Request Ambulance NOW
Nearby Hospitals
DHAKA MEDICAL
12 BEDS
1.2 km away 4.5
SQUARE HOSPITAL
2.8 km away 4.8
5 ICU
search
Bookings


CARELINK
Generate Bill
Patient:Samiya Siddiqua . BK-00123
Bed charges/day 2,000
Number of Days 3
ICU Charges 2,000
Ambulance Charges 1,000
Medicine Charges 1,200
Other Charges 500
Total 6,700
Generate Bill
Send to Patient
Dhaka Medical
Admin Panel
15 Avail Beds
4 ICU Free
15 Bookings Today
88% Occupancy
This Week
View Resouces


CARELINK
FORGOT PASSWORD
PHONE NUMBER/ EMAIL
SEND OTP
ENTER OTP
VERIFY OTP
NEW PASSWORD
CONFIRM PASSWORD
RESET PASSWORD
Go back to login page
CARELINK
CREATE ACCOUNT
NAME :
PHONE NUMBER:
EMAIL ADDRESS:
PASSWORD:
CONFRIM PASSWORD:
BLOOD GROUP:
ADDRESS:
REGISTER
ALREADY HAVE AN ACCOUNT ?
LOGIN



5. GIT WORKFLOW

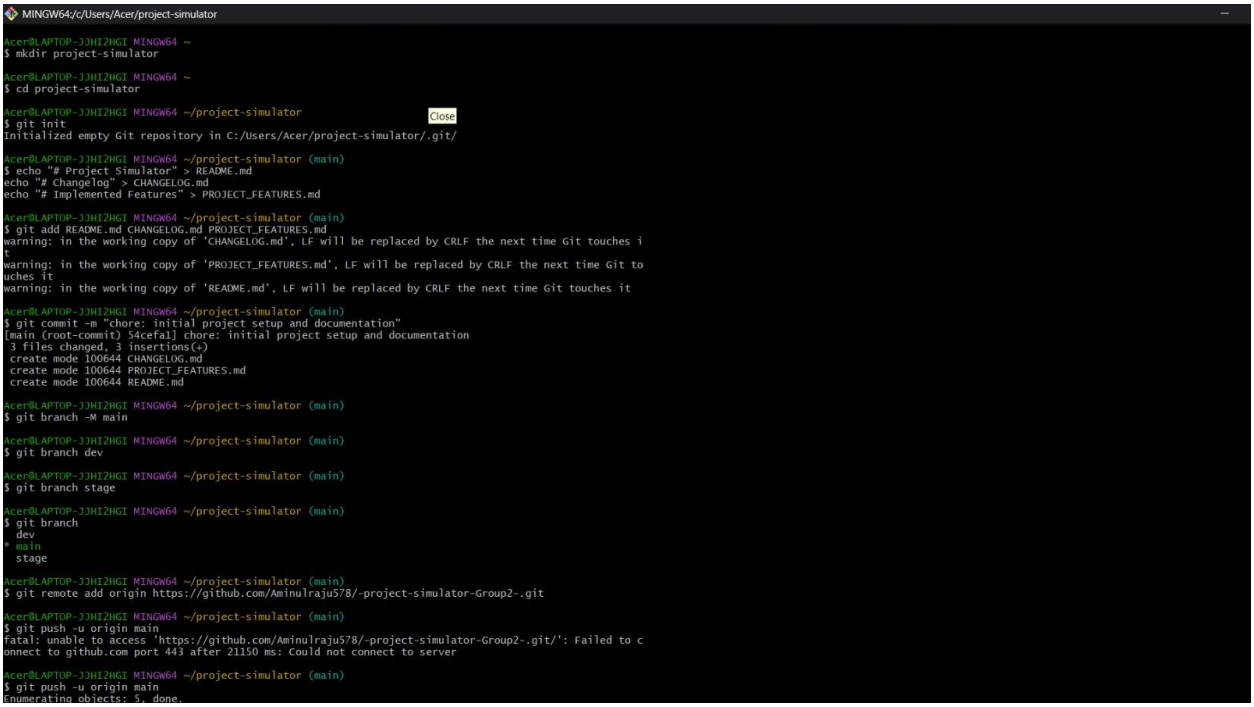
```
Acer@LAPTOP-JJHI2HGI MINGW64 ~/project-simulator (main)
$ git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (5/5), 387 bytes | 193.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Aminulraju578/-project-simulator-Group2-.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

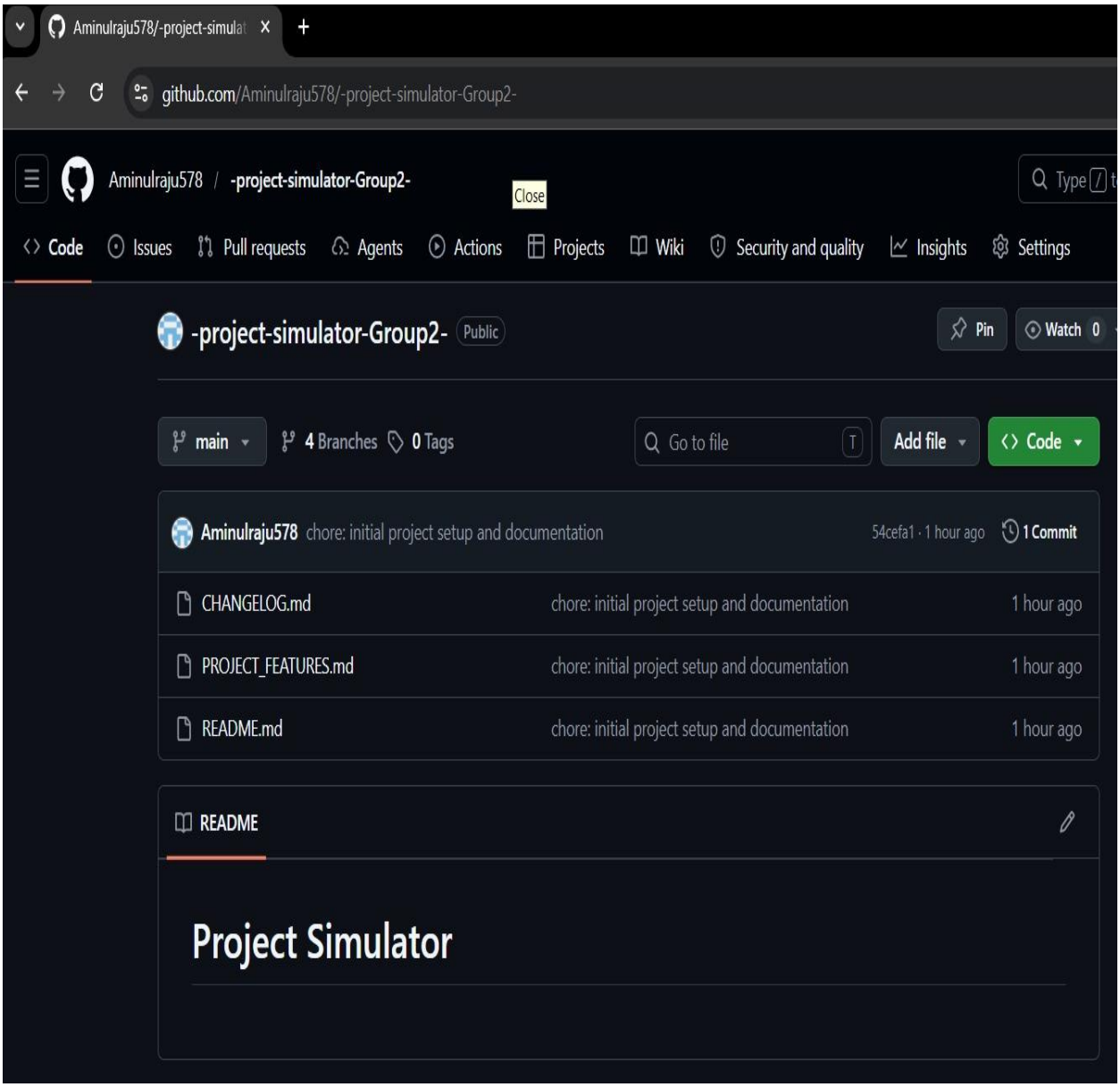
Acer@LAPTOP-JJHI2HGI MINGW64 ~/project-simulator (main)
$ git push -u origin dev
fatal: unable to access 'https://github.com/Aminulraju578/-project-simulator-Group2-.git/': Failed to c
onnect to github.com port 443 after 21388 ms: Could not connect to server

Acer@LAPTOP-JJHI2HGI MINGW64 ~/project-simulator (main)
$ git push -u origin dev
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'dev' on GitHub by visiting:
remote:      https://github.com/Aminulraju578/-project-simulator-Group2-/pull/new/dev
remote:
To https://github.com/Aminulraju578/-project-simulator-Group2-.git
 * [new branch]      dev -> dev
branch 'dev' set up to track 'origin/dev'.

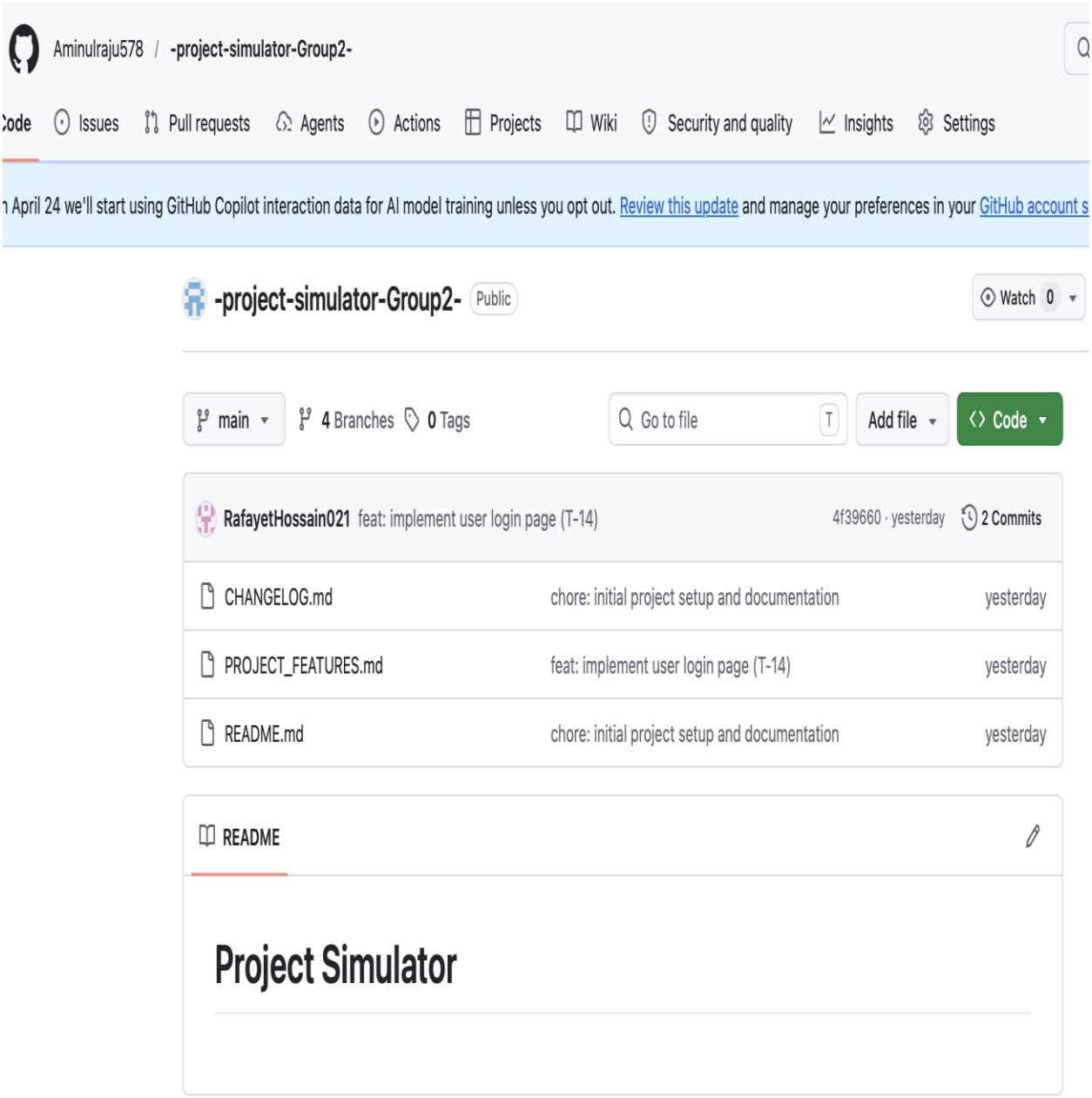
Acer@LAPTOP-JJHI2HGI MINGW64 ~/project-simulator (main)
$ git push -u origin stage
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'stage' on GitHub by visiting:
remote:      https://github.com/Aminulraju578/-project-simulator-Group2-/pull/new/stage
remote:
To https://github.com/Aminulraju578/-project-simulator-Group2-.git
 * [new branch]      stage -> stage
branch 'stage' set up to track 'origin/stage'.

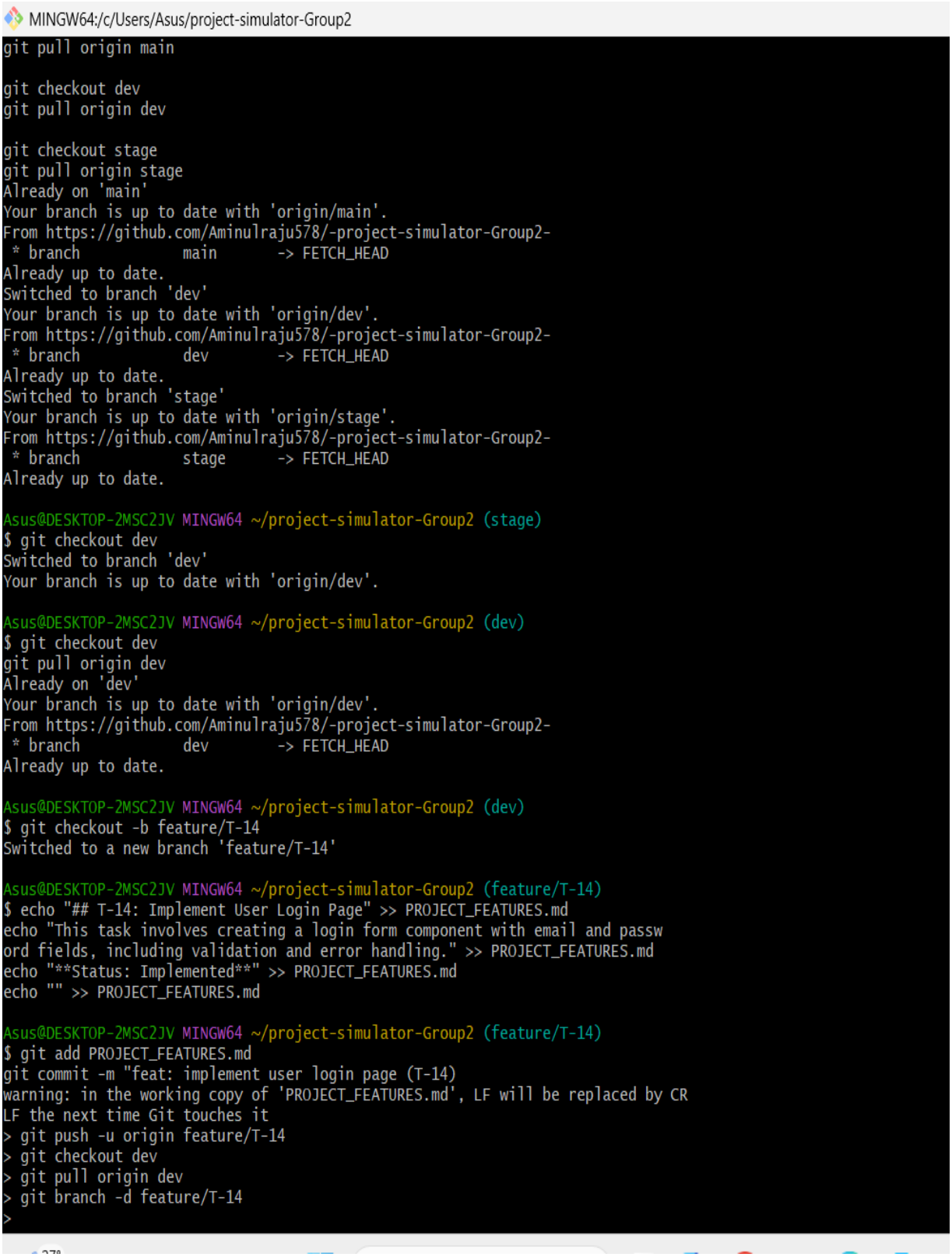
Acer@LAPTOP-JJHI2HGI MINGW64 ~/project-simulator (main)
$
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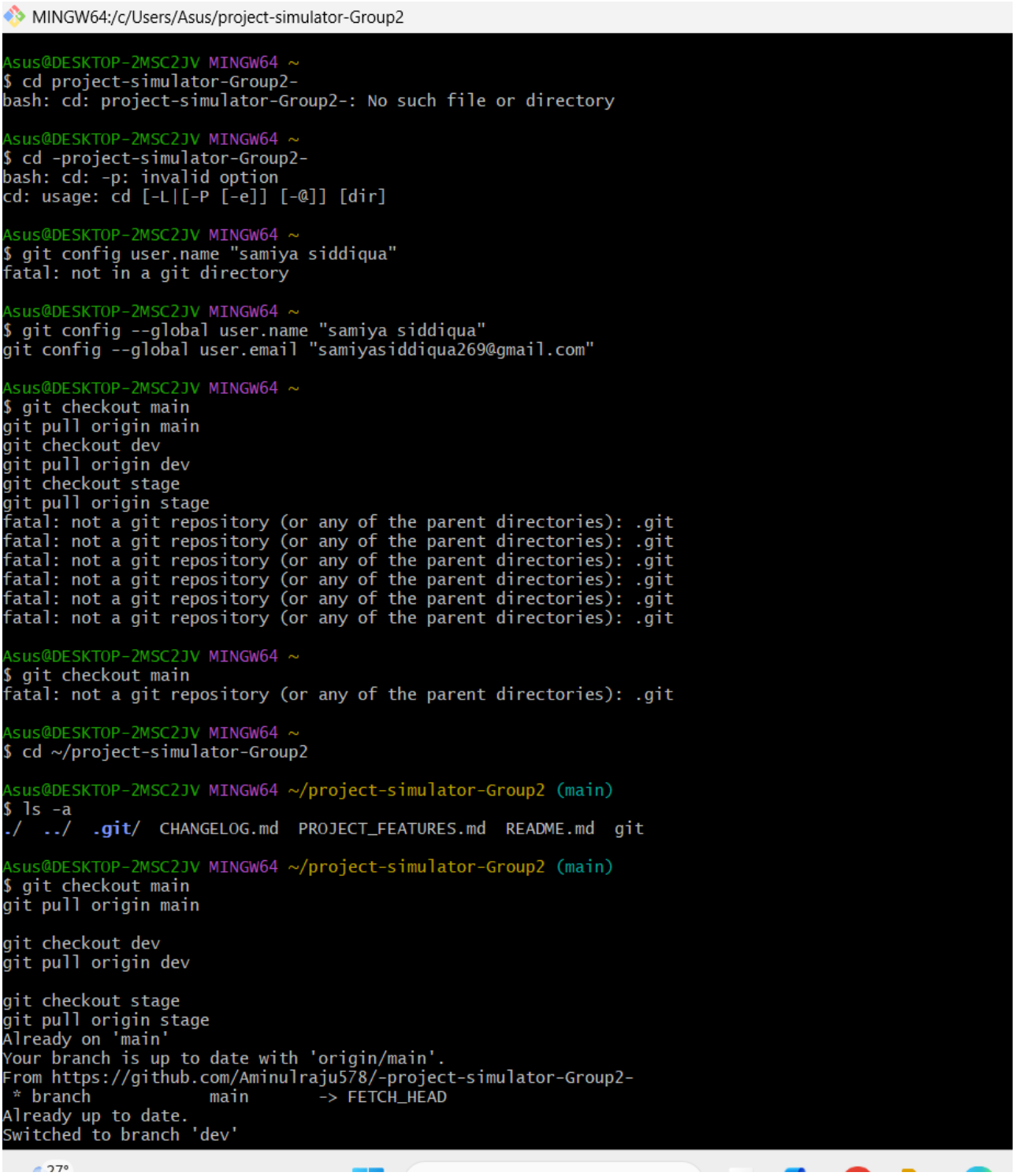


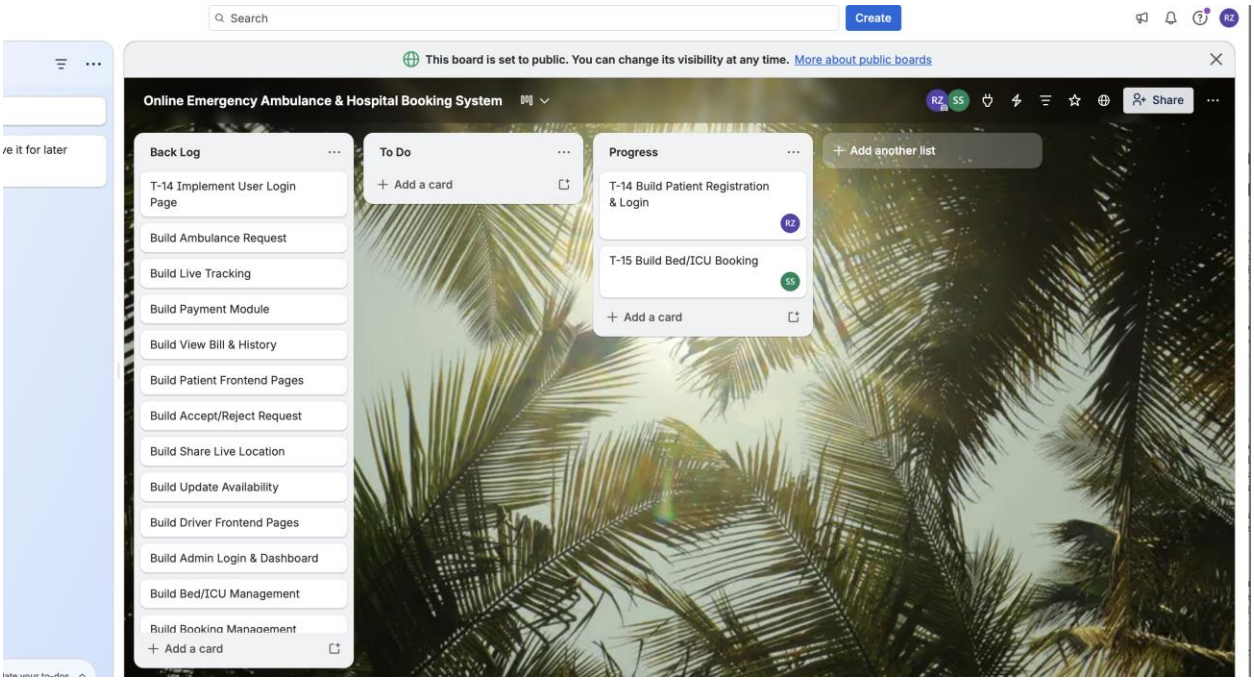

Aminulraju578/-project-simulat
github.com/Aminulraju578/-project-simulator-Group2-
Aminulraju578 / -project-simulator-Group2-
Close
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
-project-simulator-Group2-
Public
Pin
Watch 0
main
4 Branches
0 Tags
Go to file
Add file
Code
Aminulraju578 chore: initial project setup and documentation
54cefa1 · 1 hour ago
1 Commit
CHANGELOG.md
chore: initial project setup and documentation
1 hour ago
PROJECT_FEATURES.md
chore: initial project setup and documentation
1 hour ago
README.md
chore: initial project setup and documentation
1 hour ago
README
Project Simulator

```
Already up to date.
zenithmac@RHs-MacBook-Air -project-simulator-Group2- % git checkout dev
branch 'dev' set up to track 'origin/dev'.
Switched to a new branch 'dev'
zenithmac@RHs-MacBook-Air -project-simulator-Group2- % git pull origin dev
From https://github.com/Aminulraju578/-project-simulator-Group2-
 * branch            dev        -> FETCH_HEAD
Already up to date.
zenithmac@RHs-MacBook-Air -project-simulator-Group2- % git checkout stage
branch 'stage' set up to track 'origin/stage'.
Switched to a new branch 'stage'
zenithmac@RHs-MacBook-Air -project-simulator-Group2- % git pull origin stage
From https://github.com/Aminulraju578/-project-simulator-Group2-
 * branch            stage      -> FETCH_HEAD
Already up to date.
zenithmac@RHs-MacBook-Air -project-simulator-Group2- % git checkout dev
Switched to branch 'dev'
Your branch is up to date with 'origin/dev'.
zenithmac@RHs-MacBook-Air -project-simulator-Group2- % git checkout dev
Already on 'dev'
Your branch is up to date with 'origin/dev'.
zenithmac@RHs-MacBook-Air -project-simulator-Group2- % git pull origin dev
From https://github.com/Aminulraju578/-project-simulator-Group2-
 * branch            dev        -> FETCH_HEAD
Already up to date.
zenithmac@RHs-MacBook-Air -project-simulator-Group2- % git checkout -b feature/T-14
Switched to a new branch 'feature/T-14'
zenithmac@RHs-MacBook-Air -project-simulator-Group2- % echo "## T-14: Implement User Login Page" >> PROJECT_FEATURES.md
zenithmac@RHs-MacBook-Air -project-simulator-Group2- % echo "This task involves creating a login form component with email and passw
ord fields, including validation and error handling." >> PROJECT_FEATURES.md
zenithmac@RHs-MacBook-Air -project-simulator-Group2- % echo "**Status: Implemented**" >> PROJECT_FEATURES.md
zenithmac@RHs-MacBook-Air -project-simulator-Group2- % echo "" >> PROJECT_FEATURES.md
zenithmac@RHs-MacBook-Air -project-simulator-Group2- % git add PROJECT_FEATURES.md
zenithmac@RHs-MacBook-Air -project-simulator-Group2- % git commit -m "feat: implement user login page (T-14)
dquote>
dquote> '
dquote> git commit -m "feat: implement user login page (T-14)"
dquote> ''''''
dquote> "
zsh: no matches found: (T-14)\n''''''\n
zenithmac@RHs-MacBook-Air -project-simulator-Group2- % git commit -m "feat: implement user login page (T-14)"
[feature/T-14 4f39660] feat: implement user login page (T-14)
 1 file changed, 5 insertions(+)
zenithmac@RHs-MacBook-Air -project-simulator-Group2- % git push origin feature/T-14
remote: Permission to Aminulraju578/-project-simulator-Group2-.git denied to RafayetHossain021.
fatal: unable to access 'https://github.com/Aminulraju578/-project-simulator-Group2-/': The requested URL returned error: 403
zenithmac@RHs-MacBook-Air -project-simulator-Group2- % git push -u origin feature/T-14
remote: Permission to Aminulraju578/-project-simulator-Group2-.git denied to RafayetHossain021.
fatal: unable to access 'https://github.com/Aminulraju578/-project-simulator-Group2-/': The requested URL returned error: 403
zenithmac@RHs-MacBook-Air -project-simulator-Group2- % git push -u origin feature/T-14
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 497 bytes | 497.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/T-14' on GitHub by visiting:
remote:      https://github.com/Aminulraju578/-project-simulator-Group2-/pull/new/feature/T-14
remote:
To https://github.com/Aminulraju578/-project-simulator-Group2-
 * [new branch]      feature/T-14 -> feature/T-14
branch 'feature/T-14' set up to track 'origin/feature/T-14'.
zenithmac@RHs-MacBook-Air -project-simulator-Group2- % git checkout dev
Switched to branch 'dev'
Your branch is up to date with 'origin/dev'.
zenithmac@RHs-MacBook-Air -project-simulator-Group2- % git pull origin dev
From https://github.com/Aminulraju578/-project-simulator-Group2-
 * branch            dev        -> FETCH_HEAD
Already up to date.
zenithmac@RHs-MacBook-Air -project-simulator-Group2- % git branch -d feature/T-14
warning: deleting branch 'feature/T-14' that has been merged to
         'refs/remotes/origin/feature/T-14', but not yet merged to HEAD
Deleted branch feature/T-14 (was 4f39660).
zenithmac@RHs-MacBook-Air -project-simulator-Group2- %
```


Aminulraju578 / -project-simulator-Group2-
Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings
April 24 we'll start using GitHub Copilot interaction data for AI model training unless you opt out. Review this update and manage your preferences in your GitHub account s
-project-simulator-Group2- Public
Watch 0
main 4 Branches 0 Tags
Go to file T
Add file
Code
RafayetHossain021 feat: implement user login page (T-14)
4f39660 · yesterday
2 Commits
CHANGELOG.md chore: initial project setup and documentation yesterday
PROJECT_FEATURES.md feat: implement user login page (T-14) yesterday
README.md chore: initial project setup and documentation yesterday
README
Project Simulator


MINGW64:/c/Users/Asus/project-simulator-Group2
git pull origin main
git checkout dev
git pull origin dev
git checkout stage
git pull origin stage
Already on 'main'
Your branch is up to date with 'origin/main'.
From https://github.com/Aminulraju578/-project-simulator-Group2-
* branch main -> FETCH_HEAD
Already up to date.
Switched to branch 'dev'
Your branch is up to date with 'origin/dev'.
From https://github.com/Aminulraju578/-project-simulator-Group2-
* branch dev -> FETCH_HEAD
Already up to date.
Switched to branch 'stage'
Your branch is up to date with 'origin/stage'.
From https://github.com/Aminulraju578/-project-simulator-Group2-
* branch stage -> FETCH_HEAD
Already up to date.
Asus@DESKTOP-2MSC2JV MINGW64 ~/project-simulator-Group2 (stage)
$ git checkout dev
Switched to branch 'dev'
Your branch is up to date with 'origin/dev'.
Asus@DESKTOP-2MSC2JV MINGW64 ~/project-simulator-Group2 (dev)
$ git checkout dev
git pull origin dev
Already on 'dev'
Your branch is up to date with 'origin/dev'.
From https://github.com/Aminulraju578/-project-simulator-Group2-
* branch dev -> FETCH_HEAD
Already up to date.
Asus@DESKTOP-2MSC2JV MINGW64 ~/project-simulator-Group2 (dev)
$ git checkout -b feature/T-14
Switched to a new branch 'feature/T-14'
Asus@DESKTOP-2MSC2JV MINGW64 ~/project-simulator-Group2 (feature/T-14)
$ echo "## T-14: Implement User Login Page" >> PROJECT_FEATURES.md
echo "This task involves creating a login form component with email and passw
ord fields, including validation and error handling." >> PROJECT_FEATURES.md
echo "**Status: Implemented**" >> PROJECT_FEATURES.md
echo "" >> PROJECT_FEATURES.md
Asus@DESKTOP-2MSC2JV MINGW64 ~/project-simulator-Group2 (feature/T-14)
$ git add PROJECT_FEATURES.md
git commit -m "feat: implement user login page (T-14)
warning: in the working copy of 'PROJECT_FEATURES.md', LF will be replaced by CR
LF the next time Git touches it
> git push -u origin feature/T-14
> git checkout dev
> git pull origin dev
> git branch -d feature/T-14
>


MINGW64:/c/Users/Asus/project-simulator-Group2
Asus@DESKTOP-2MSC2JV MINGW64 ~
$ cd project-simulator-Group2-
bash: cd: project-simulator-Group2-: No such file or directory
Asus@DESKTOP-2MSC2JV MINGW64 ~
$ cd -project-simulator-Group2-
bash: cd: -p: invalid option
cd: usage: cd [-L|[-P [-e]] [-@]] [dir]
Asus@DESKTOP-2MSC2JV MINGW64 ~
$ git config user.name "samiya siddiqua"
fatal: not in a git directory
Asus@DESKTOP-2MSC2JV MINGW64 ~
$ git config --global user.name "samiya siddiqua"
git config --global user.email "samiyasiddiqua269@gmail.com"
Asus@DESKTOP-2MSC2JV MINGW64 ~
$ git checkout main
git pull origin main
git checkout dev
git pull origin dev
git checkout stage
git pull origin stage
fatal: not a git repository (or any of the parent directories): .git
fatal: not a git repository (or any of the parent directories): .git
fatal: not a git repository (or any of the parent directories): .git
fatal: not a git repository (or any of the parent directories): .git
fatal: not a git repository (or any of the parent directories): .git
fatal: not a git repository (or any of the parent directories): .git
Asus@DESKTOP-2MSC2JV MINGW64 ~
$ git checkout main
fatal: not a git repository (or any of the parent directories): .git
Asus@DESKTOP-2MSC2JV MINGW64 ~
$ cd ~/project-simulator-Group2
Asus@DESKTOP-2MSC2JV MINGW64 ~/project-simulator-Group2 (main)
$ ls -a
./ ../ .git/ CHANGELOG.md PROJECT_FEATURES.md README.md git
Asus@DESKTOP-2MSC2JV MINGW64 ~/project-simulator-Group2 (main)
$ git checkout main
git pull origin main
git checkout dev
git pull origin dev
git checkout stage
git pull origin stage
Already on 'main'
Your branch is up to date with 'origin/main'.
From https://github.com/Aminulraju578/-project-simulator-Group2-
* branch main -> FETCH_HEAD
Already up to date.
Switched to branch 'dev'



6. SOFTWARE TESTING

The following testing methods were identified and applied during the testing phase of the CareLink - Online Emergency Ambulance and Hospital Booking System:

Unit Testing

Unit testing was performed to test individual components and functions of the system in isolation. Each module such as user registration, bed booking, ambulance request, and payment processing was tested independently to verify that eac Create h function produces the correct output for a given input.

Integration Testing

Integration testing was conducted to verify that different modules of the system work correctly together. For example, the interaction between the Patient module and the Hospital module was tested to ensure that booking a bed correctly updates the hospital's availability count. Similarly the connection between the Ambulance Request module and the Driver module was tested to ensure requests are received and processed correctly.

Black Box Testing

Black box testing was used to test the system based on its inputs and expected outputs without looking at the internal code. Test cases were prepared based on the functional requirements defined in the SRS document. Testers entered various inputs through the user interface and verified whether the outputs matched the expected results.

Manual Testing

Manual test cases were prepared for all major features of the system. Each test case includes the test steps, test data, expected results, actual results, and pass or fail status. A total of 8 manual test cases were prepared by the four team members, with 2 test cases per person.

Test Case 1

Project Name: Online Emergency Ambulance & Hospital Booking System			**Test Designed by: Ishrak Jahan Mim**	
Test Case ID: TC_1			**Test Designed date: 25/04/2026**	
Test Priority (Low, Medium, High): High			**Test Executed by: Ishrak Jahan Mim**	
Module Name: Patient Login			**Test Execution date: 28/04/2026**	
Test Title: Verify Patient Login with Valid Username and Password				
Description: Test the patient login page of the Online Emergency Ambulance & Hospital Booking System				
Precondition: The patient has a registered account with a valid phone number and password				
Dependencies: Patient Registration must be completed before this test				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. **Go to the system login page**	Phone: 01712345678	The patient should be logged in	As expected	Pass

2. **Select role as Patient** 3. **Enter valid phone number** 4. **Enter valid password** 5. **Click the Login button**	Password: Patient@123	successfully and redirected to the Patient Dashboard		

Test Case 2

Project Name: Online Emergency Ambulance & Hospital Booking System			**Test Designed by: MD.Rafayet Hossain Zanith**	
Test Case ID: TC_2			**Test Designed date: 24/04/2026**	
Test Priority (Low, Medium, High): High			**Test Executed by: MD.Rafayet Hossain Zanith**	
Module Name: Bed / ICU Booking			**Test Execution date: 28/04/2026**	
Test Title: Verify Patient Can Book an Available Hospital Bed Successfully				
Description: Test the hospital bed booking feature for a logged-in patient in the system				
Precondition: The patient is logged in. At least one hospital has available beds in the system				
Dependencies: TC_1 (Patient Login) must pass before this test				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. **Login as Patient**	Phone: 01712345678	The booking should be confirmed with a unique	As expected	Pass

2. **Click on Search Nearby Hospitals** 3. **Select a hospital from the list** 4. **Click on Check Bed / ICU Availability** 5. **Select bed type as General Bed** 6. **Click Book Now button** 7. **Confirm the booking**	Password: Patient@123 Hospital: Dhaka Medical Bed Type: General Bed	booking ID generated and bed availability count decreased by 1		

Test Case 3

Project Name: Online Emergency Ambulance & Hospital Booking System	**Test Designed by: SAMIYA SIDDIQUA SYMPA**
Test Case ID: TC_3	**Test Designed date: 23/04/2026**
Test Priority (Low, Medium, High): High	**Test Executed by: SAMIYA SIDDIQUA SYMPA**
Module Name: Ambulance Request	**Test Execution date: 28/04/2026**
Test Title: Verify Patient Can Successfully Request an Ambulance	
Description: Test the ambulance request feature where a patient sends an emergency ambulance request from their location	
Precondition: Patient is logged in. At least one ambulance driver is available and online in the system	

Dependencies: TC_1 (Patient Login) must pass before this test				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. **Login as Patient** 2. **Click on Request Ambulance** 3. **Allow GPS location access** 4. **Enter pickup address manually if needed** 5. **Select destination hospital** 6. **Click Send Request button** 7. **Wait for driver confirmation**	Phone: 01712345678 Password: Patient@123 Pickup: Dhanmondi, Dhaka Hospital: Square Hospital	Ambulancerequest should be sent successfully. Patient should see a confirmation message and request status should change to Pending	As expected	Pass

Test Case 4

Project Name: Online Emergency Ambulance & Hospital Booking System	**Test Designed by: AMINUL ISLAM RAJU**
Test Case ID: TC_4	**Test Designed date: 26/04/2026**
Test Priority (Low, Medium, High): High	**Test Executed by: AMINUL ISLAM RAJU**
Module Name: Ambulance Driver - Accept Request	**Test Execution date: 28/04/2026**

Test Title: Verify Ambulance Driver Can Accept an Incoming Patient Request				
Description: Test that the ambulance driver receives the patient request notification and can accept it successfully				
Precondition: Driver is logged in and status is set to Online. A patient has submitted an ambulance request				
Dependencies: TC_3 (Ambulance Request) must pass before this test				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. **Login as Ambulance Driver** 2. **Set availability status to Online** 3. **Wait for incoming request notification** 4. **Click on View Incoming Requests** 5. **Review patient pickup location** 6. **Click Accept Request button**	Phone: 01898765432 Password: Driver@123 Status: Online	Driver should successfully accept the request. Request status should update to Accepted. Patient should receive a notification that driver is on the way	As expected	Pass

Test Case 5

Project Name: Online Emergency Ambulance & Hospital Booking System	Test Designed by: Ishrak Jahan Mim

Test Case ID: TC_5			**Test Designed date: 25/04/2026**	
Test Priority (Low, Medium, High): High			**Test Executed by: Ishrak Jahan Mim**	
Module Name: Online Payment			**Test Execution date: 28/04/2026**	
Test Title: Verify Patient Can Make Online Payment for Hospital Booking				
Description: Test the online payment feature where a patient pays for a confirmed hospital bed booking using mobile banking				
Precondition: Patient is logged in and has a confirmed hospital booking. Payment gateway is connected				
Dependencies: TC_2 (Bed Booking) must pass before this test				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. **Login as Patient** 2. **Go to My Bookings** 3. **Select the confirmed booking** 4. **Click on Make Payment** 5. **Select payment method as Mobile Banking** 6. **Enter mobile banking number** 7. **Enter OTP received on phone**	Phone: 01712345678 Password: Patient@123 Booking ID: BK-00123 Method: bKas bKash No: 01712345678 OTP: 456789	Payment should be processed successfully. A payment receipt should be generated with transaction ID. Booking status should update to Paid	As expected	Pass

8. **Click Confirm Payment**				

Test Case 6

Project Name: Online Emergency Ambulance & Hospital Booking System			**Test Designed by: MD.Rafayet Hossain Zanith**	
Test Case ID: TC_6			**Test Designed date: 25/04/2026**	
Test Priority (Low, Medium, High): Medium			**Test Executed by: MD.Rafayet Hossain Zanith**	
Module Name: Hospital Admin - Update Bed Availability			**Test Execution date: 28/04/2026**	
Test Title: Verify Hospital Admin Can Update Bed and ICU Availability				
Description: Test that the hospital admin can update the real-time bed and ICU availability numbers in the system				
Precondition: Hospital Admin is logged in with valid admin credentials and has access to the hospital management dashboard				
Dependencies: Admin account must be created and verified before this test				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. **Login as Hospital Admin** 2. **Go to Admin Dashboard**	Email: admin@dhakamed.com Password: Admin@123 General Beds: 15	Bed availability should be updated successfully. The new numbers	As expected	Pass

3. **Click on Manage Bed Availability** 4. **Update available General Beds count** 5. **Update available ICU Beds count** 6. **Update available Cabin count** 7. **Click Save Changes button**	ICU Beds: 5 Cabins: 8	should reflect immediately on the patient-facing hospital search page in real time		

Test Case 7

Project Name: Online Emergency Ambulance & Hospital Booking System	**Test Designed by: SAMIYA SIDDIQUA SYMPA**
Test Case ID: TC_7	**Test Designed date: 24/04/2026**
Test Priority (Low, Medium, High): Medium	**Test Executed by: SAMIYA SIDDIQUA SYMPA**
Module Name: Patient - Cancel Booking	**Test Execution date: 28/04/2026**
Test Title: Verify Patient Can Cancel an Existing Hospital Booking	
Description: Test that the patient can cancel a previously confirmed hospital bed booking and the bed count is restored	

Precondition: Patient is logged in and has at least one active confirmed booking in the system				
Dependencies: TC_2 (Bed Booking) must pass before this test				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. **Login as Patient** 2. **Go to View Booking History** 3. **Select an active booking** 4. **Click on Cancel Booking button** 5. **A confirmation prompt appears** 6. **Click Yes to confirm cancellation**	Phone: 01712345678 Password: Patient@123 Booking ID: BK-00123	Booking should be cancelled successfully. Booking status should change to Cancelled. Bed availability count should increase by 1. Patient should receive a cancellation notification	As expected	Pass

Test Case 8

Project Name: Online Emergency Ambulance & Hospital Booking System	**Test Designed by: AMINUL ISLAM RAJU**
Test Case ID: TC_8	**Test Designed date: 25/04/2026**
Test Priority (Low, Medium, High): High	**Test Executed by: AMINUL ISLAM RAJU**
Module Name: Patient Login - Invalid Credentials	**Test Execution date: 28/04/2026**

Test Title: Verify System Rejects Login with Invalid Password				
Description: Test that the system does not allow a patient to login with a wrong password and displays an appropriate error message				
Precondition: The patient has a registered account in the system				
Dependencies: Patient Registration must be completed before this test				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. **Go to the system login page** 2. **Select role as Patient** 3. **Enter valid registered phone number** 4. **Enter incorrect password** 5. **Click the Login button**	Phone: 01712345678 Password: WrongPass@999	The system should not allow login. An error message should be displayed saying Invalid phone number or password. The patient should remain on the login page	As expected	Pass

7. CONCLUSION

The CareLink - Online Emergency Ambulance and Hospital Booking System is a comprehensive digital healthcare platform designed to address the critical challenges faced by patients, ambulance drivers, and hospital administrators in Bangladesh during emergency situations. The system provides a centralized solution for real-time hospital resource management, seamless bed and ICU booking, on-demand ambulance services, and live tracking.

Throughout the development of this project, the team successfully completed all major phases including requirements gathering, UI/UX design, system design, and testing. The system was

designed with three clearly defined user roles, each with their own set of functional requirements and interface designs. A total of 28 Figma screens were designed to represent the complete user interface, and 28 use cases were identified and modeled in the use case diagram with proper include and extend relationships.

The project followed the Agile development model which allowed the team to work iteratively and adapt to changes efficiently. Version control was maintained using GitHub with feature branches for each module, ensuring clean and organized collaboration among all team members. The testing phase covered all major functionalities through 8 detailed manual test cases, all of which passed successfully.

This project demonstrates how technology can be used to solve real-world healthcare problems and improve emergency response times. With further development and deployment, this system has the potential to significantly reduce the time required for patients to access emergency medical care in Bangladesh and other developing countries.